Energy-Efficient Scheduling for Overloaded RSMA MU-MIMO Uplink under Per-SNR Delivery Constraints: Hierarchical Reinforcement Learning with Solver-Expert Distillation

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***Abstract*—Rate-splitting multiple access (RSMA) promises spectral gains in the *overloaded* multi-user MIMO uplink, where the number of users K exceeds the receive antennas N_r, but harvesting those gains requires jointly choosing transmit power, the common/private stream split, and the successive-interference-cancellation (SIC) decoding order under imperfect channel state information (CSI) and hard per-user quality-of-service (QoS) limits. We formulate energy-efficiency (EE) maximization under *per-SNR-point* delivery-ratio and queue-stability constraints relative to a strong SDMA reference—constraints that must hold *simultaneously at every operating point*, not on average—and show that this requirement defeats both classical baselines and single-timescale deep reinforcement learning (DRL). We propose a two-timescale hierarchical policy: a slow meta-controller that sets user grouping and group-level stream split, and a fast self-attention graph worker that proposes per-slot power shares, queue drain ratios, and a Plackett-Luce SIC order. The worker is trained with a delivery-gated Lyapunov composite reward (DGLCR) combining a Dinkelbach-linearized EE term, max-weight throughput, and CVaR-tail delivery-virtual-queue penalties, under a Lagrangian delivery constraint with lexicographic gradient projection. Because the reward landscape is non-differentiable through the combinatorial SIC order, we introduce a *solver expert*: a differentiable shadow of the MMSE-SIC rate engine that refines power logits online and supplies demonstrations; the policy is distilled by behavior cloning and fine-tuned with a stepwise-frozen protocol (SFTP) with guard-selected checkpoints. On a 7-point SNR grid (0–30 dB, imperfect CSI, 50 episodes/point), the policy clears all 7 per-point gates with mean delivery 0.9496 ± 0.0047 across five seed replications and dominates SDMA, NOMA and equal-power baselines on delivery, queue and overflow at every SNR—at roughly half their transmit power (30 dB: 0.888 vs. 0.664/0.581/0.601). We further show that the solver expert *itself*, run as a training-free per-slot optimizer, is the strongest controller on this grid (mean delivery 0.9574), establishing a model-based upper bound; distillation compresses it into a single-forward-pass policy (0.9441) and the frozen fine-tune recovers ≈59% of the compression gap (0.9519), so the learned policy's value is inference cost and model-free deployment rather than absolute performance. A pre-registered fine-tuning campaign further shows that released-learning-rate, trust-region, anchoring, and data-reshaping variants all fall within the policy's own seed dispersion, motivating multi-seed confidence-interval comparison as the honest reporting standard for this class of methods. Finally, a Shapley-value-based interpretability analysis (global feature-subset selection via the SVEA classification game, plus expected-gradient attributions validated against a model-agnostic KernelSHAP proxy under multiple reference constructions) shows that the learned scheduler's power decisions are driven foremost by queue backlog and channel gain, with the queue block as the dominant feature across operating points.**

***Index Terms*—Rate-splitting multiple access, MU-MIMO uplink, energy efficiency, deep reinforcement learning, MMSE-SIC, Lyapunov optimization, fractional programming, expert distillation.**

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I. Introduction

Cell-edge and machine-type traffic increasingly force base stations to serve more active users than they have receive antennas. In this *overloaded regime* ($K > N_r$) spatial multiplexing alone saturates: interference, not noise, bounds the sum rate, and queueing delays grow without a scheduling policy that is explicitly queue-aware. Rate-splitting multiple access (RSMA) mitigates overload by splitting each message into a *common* stream decoded by all users and a *private* stream decoded by its intended user, with successive interference cancellation (MMSE-SIC) at the receiver [1]–[3]. Realizing RSMA's promise, however, requires solving a coupled, per-slot combinatorial problem—power allocation across users, the common/private split, and the SIC decoding order—repeatedly and online, under imperfect CSI and a realistic power-consumption model.

Three properties make this problem resistant to existing approaches. *(1) Hard per-point constraints:* practical deployments specify QoS at discrete operating points (SNR levels), e.g. a minimum delivery ratio and a maximum queue length *at each point*, benchmarked against a deployed baseline such as SDMA; a policy excellent on average but violating one point is unacceptable. *(2) Imperfect CSI in the overloaded regime:* pilot-based estimation injects noise whose variance scales as $\sigma^2/(Kp_{\text{pilot}})$ per user, so overload worsens estimation quality exactly where spatial resources are scarcest; robust designs bound the residual error $\|\Delta\mathbf{h}_k\| \le \varepsilon$ but the interaction with SIC error propagation makes closed-form robust allocation brittle. *(3) Energy efficiency without loopholes:* maximizing $\text{EE} = R_{\text{sum}}/P_{\text{consume}}$ under $P_{\text{consume}} = \sum_k P_k/\eta_{\text{PA}} + P_{\text{circ}}$ removes the degenerate "transmit nothing" optimum but couples the fractional objective to the queueing state, requiring fractional-programming treatment (Dinkelbach) inside a stochastic control loop.

Deep reinforcement learning is a natural candidate for (1)–(3), yet naive DRL fails for a structural reason: the combina-

torial SIC order and non-differentiable rate expressions block gradient-based credit assignment, and the reward's delivery gate creates sparse, saturating gradients at exactly the points that matter. Recent robust and distributionally-robust DRL beamforming [4]–[6] and generalization analyses [7], [8] address robustness to CSI perturbation but not the simultaneous per-point QoS structure. Lyapunov drift-plus-penalty methods [9]–[11] provide queue-stability guarantees for *given* rate regions but do not learn the combinatorial allocation itself; policy-distillation practice in RANs [12] motivates our teacher→student design but does not couple distillation to per-slot combinatorial refinement under hard per-point gates.

Related lines of work span: expert-driven imitation and expert iteration [23]–[27]; deep-RL foundations for continuous control and discrete optimisers [28]–[33]; queue-stability and cross-layer scheduling theory [34], [35]; RSMA and NOMA rate-region analysis [3], [36], [37]; deep-learning and deep-RL resource allocation in wireless systems [38]–[40]; and fractional-programming tools for communication EE [15], [41].

Contributions. (i) We formalize EE maximization for overloaded RSMA MU-MIMO uplink under per-SNR-point delivery and queue constraints relative to SDMA, with a lexicographic criterion isolating the binding constraints. (ii) A two-timescale hierarchical policy: slow meta-controller (grouping + group-level split) and fast self-attention graph worker (power shares, drain ratios, Plackett-Luce SIC order) with state-augmented observations. (iii) A delivery-gated Lyapunov composite reward (DGLCR) with Dinkelbach linearization, max-weight throughput and CVaR-tail delivery-virtual-queue penalties, under a Lagrangian delivery constraint with lexicographic gradient projection. (iv) Solver-expert distillation: a differentiable shadow of the MMSE-SIC engine (float64-consistent to 10^{-9}) that refines power logits online, supplies demonstrations that alone clear all 7 gates, and is distilled into the policy; a stepwise-frozen fine-tune protocol (SFTP) with guard-selected checkpoints preserves and extends the distilled behavior. (v) An honest empirical account: full-grid comparison against SDMA/NOMA/equal-power; seed replication quantifying intrinsic dispersion ($\sigma_{\text{seed}} \approx 0.005$); and a pre-registered fine-tuning campaign showing five mechanism variants fall within that dispersion. (vi) A Shapley-value-based interpretability analysis (§V-J): global feature-subset selection via the SVEA classification game and expected-gradient attributions validated against a model-agnostic proxy under multiple references, showing that the learned power decision is driven by queue backlog and channel gain while the drain, decoding order and meta-grouping heads collapse to near-constant defaults.

II. System Model and Problem Formulation

A. Network and RSMA transmission

A single base station with N_r antennas serves K users, each with N_t transmit antennas, over T slots per episode. User k's message is split into a common part (share governed by $\alpha_k \in [\alpha_{\min}, \alpha_{\max}]$) and a private part. Users assigned to group g share a common stream decoded by all group members; all common streams precede all private streams in the decoding order, and common-stream SIC cancellation is performed only within a group. The transmitted signal of user k is

$$x_k = \sqrt{\alpha_k p_k}\, s_k^c + \sqrt{(1-\alpha_k)p_k}\, s_k^p, \qquad \mathbb{E}|s_k^c|^2 = \mathbb{E}|s_k^p|^2 = 1, \tag{1}$$

where $\alpha_k = 0$ recovers unsplit (SDMA/NOMA-type) transmission. The BS applies unified MRT transmit weights $\mathbf{W}_c, \mathbf{W}_p \in \mathbb{C}^{N_r \times K}$ under per-user caps $P_k \le P_{k,\max}$. The receiver performs MMSE-SIC in decoding order π. For a stream decoded at position j in π, the rate is $R = \log_2(1 + \text{SINR})$ with SINR denominator collecting (i) interference from not-yet-decoded streams, (ii) SIC residual from decoded streams scaled by hardware imperfection β^2, and (iii) CSI-error residual scaled by $\varepsilon^2 p$, plus thermal noise σ^2:

$$\begin{aligned} \text{SINR}_k^{(j)} &= \frac{p_k\, |\mathbf{w}^{\mathsf{H}} \hat{\mathbf{h}}_k|^2}{I_{\text{undec}} + \beta^2 I_{\text{dec}} + \varepsilon^2 P_{\text{tot}} \|\mathbf{w}\|^2 + \sigma^2 \|\mathbf{w}\|^2}, \\ I_{\text{undec}} &= \sum_{i \notin \mathcal{D}_j} p_i\, |\mathbf{w}^{\mathsf{H}} \hat{\mathbf{h}}_i|^2, \qquad I_{\text{dec}} = \sum_{i \in \mathcal{D}_j} p_i\, |\mathbf{w}^{\mathsf{H}} \hat{\mathbf{h}}_i|^2, \end{aligned} \tag{2}$$

where $\mathcal{D}_j$ is the set decoded before position j. The MMSE combiner is regularized by the worst-case bound via $\varepsilon^2 P_{\text{tot}} \mathbf{I}$ (robust MMSE). Evaluating (2) at the common- and private-stream positions of user k gives $\text{SINR}_k^c, \text{SINR}_k^p$, and the per-user and sum rates follow as

$$\begin{aligned} R_k &= \log_2\big(1 + \text{SINR}_k^c\big) + \log_2\big(1 + \text{SINR}_k^p\big), \\ R_{\text{sum}} &= \sum_{k=1}^{K} R_k. \end{aligned} \tag{3}$$

Declared rates use $(\hat{\mathbf{H}}, \varepsilon)$; actual rates use $(\mathbf{H}, \varepsilon{=}0)$; the outage-aware rate is their minimum, so over-declared rates are penalized by real throughput.

B. Imperfect CSI and channel dynamics

Channels follow a 3GPP UMi path-loss model (users at 50–300 m, exponent 3.76, normalized at 100 m) with AR(1) temporal correlation

$$\mathbf{h}_k[t] = \rho\, \mathbf{h}_k[t-1] + \sqrt{1-\rho^2}\, \Delta\mathbf{h}_k[t], \tag{4}$$

$$\Delta\mathbf{h}_k[t] \sim \mathcal{CN}\big(\mathbf{0},\, \sigma_h^2 \mathbf{I}\big), \quad \rho = 0.9, \tag{5}$$

i.e. slow Rayleigh fading whose coherence grows with ρ. Pilot-based MMSE estimation yields per-user error variance

$$\sigma_{\text{est}}^2 = \frac{\sigma^2}{K\, p_{\text{pilot}}} + \varepsilon_{\text{floor}}^2, \tag{6}$$

so estimation quality degrades with load K; the residual is bounded $\|\Delta\mathbf{h}_k\| \le \varepsilon$. Training randomizes noise power over $\sigma^2 \in [-5, 35]$ dB per episode; evaluation is at fixed points under imperfect CSI.

C. Traffic, queues and QoS classes

Arrivals $A_k[t]$ are Poisson with SNR-adaptive mean $\lambda(\text{SNR}) = \text{load}(\text{SNR})\, C(\text{SNR})/K$, $\text{load}(\cdot)$ a sigmoid from

0.75 to 1.3; three QoS classes scale arrival/drift/overflow weights. Queues evolve as

$$Q_k[t{+}1] = \min\{\max\{Q_k[t] - D_k[t], 0\} + A_k[t],\, Q_{\max}\}, \quad (7)$$

with served bits

$$D_k[t] = \min\{Q_k[t],\ d_k[t]\, R_k[t]\}, \qquad d_k[t] \in [0,1], \quad (8)$$

where d_k is the worker's queue-drain ratio, and overflow $O_k[t]$ counting bits dropped at $Q_{\max}$. The delivery ratio over an episode is

$$\text{deliv} = \frac{\sum_k \sum_t D_k[t]}{\sum_k \sum_t A_k[t]}, \quad (9)$$

the fraction of arrived bits served within the horizon.

D. Power consumption and energy efficiency

$$P_{\text{consume}} = \frac{\sum_k P_k}{\eta_{\text{PA}}} + P_{\text{circ}}, \quad \text{EE} = \frac{R_{\text{sum}}}{P_{\text{consume}}}, \quad (10)$$

with $\eta_{\text{PA}} = 0.35$, $P_{\text{circ}} = 5.0\,\text{W}$ and a floor $P_{\text{eff}} = \max(P_{\text{consume}}, P_{\text{floor}})$ removing the degenerate $P \to 0$ optimum.

E. Problem formulation and evaluation criterion

Let $\mathcal{S} = \{s_1, \ldots, s_7\}$ be the evaluation SNR grid (0–30 dB). The constraint set, relative to a reference SDMA policy, is

$$\begin{aligned} &\text{deliv}_\pi(s) \geq \text{deliv}_{\text{SDMA}}(s), \\ &\text{queue}_\pi(s) \leq 1.1\, \text{queue}_{\text{SDMA}}(s), \end{aligned} \quad (11)$$

for all $s \in \mathcal{S}$ (a *relative two-bar gate* at each point). The reported score is the lexicographic tuple

$$\text{score} = \big[\ \#\{s : \text{gate met}\},\ \overline{\text{deliv}},\ -\overline{\text{queue}},\ \text{EE ratio}\ \big], \quad (12)$$

so cleared points dominate first, then mean delivery, queue, and EE ratio. A global aggregate check gates termination but does not substitute (11). Collecting the per-slot decisions into $\mathbf{x}[t] = (\mathbf{g}, \alpha, \mathbf{p}, \mathbf{d}, \pi)[t]$, the design problem is the constrained stochastic control problem

$$\begin{aligned} &\max_{\{\mathbf{x}[t]\}} \ \liminf_{T\to\infty} \ \frac{1}{T} \sum_{t=0}^{T-1} \text{EE}[t] \\ &\text{s.t.} \quad (11), \quad p_k[t] \leq P_{k,\max}, \quad \bar{Q}_k < \infty, \quad \pi[t] \in \mathcal{S}_K, \end{aligned} \quad (13)$$

where $\mathcal{S}_K$ denotes the valid SIC permutations. Because a single policy must satisfy (11) at *every* point simultaneously, (13) cannot be decomposed per SNR point, which is precisely the structure that defeats per-point-agnostic baselines (§V).

III. Proposed Framework

A. Two-timescale MDP

The slow scale (every $T_s = 20$ slots) selects grouping $g_k \in \{0, \ldots, G-1\}$ and group-level split $\alpha \in [\alpha_{\min}, \alpha_{\max}]^G$ expanded per user. The fast scale (every slot) selects the $4K$-dim action: power-share logits (softmax shares $\times K$, clipped), drain ratio $d_k \in [0,1]$, and Plackett-Luce scores z_k [13], [14] inducing SIC order π by Gumbel top-K sampling:

$$\begin{aligned} &P(\pi \mid \mathbf{z}) = \prod_{i=1}^{K} \frac{z_{\pi(i)}}{\sum_{j=i}^{K} z_{\pi(j)}}, \\ &\pi = \arg\text{sort}\big(-(\log \mathbf{z} + \gamma)\big), \quad \gamma_i \sim \text{Gumbel}(0,1), \end{aligned} \quad (14)$$

with $\log P(\pi \mid \mathbf{z}) = \sum_i \big[\log z_{\pi(i)} - \log \sum_{j\geq i} z_{\pi(j)}\big]$ supplying the differentiable permutation log-probability for the policy gradient; at deployment the order is deterministic, $\pi = \arg\text{sort}(-\mathbf{z})$. Power logits are mapped to shares by a clipped softmax over users, so the $4K$ action stays in a bounded domain. Observations are per-user feature blocks plus globals; the worker observation is $13K{+}3$ dimensional — per-user channel gain, queue, queue change, arrivals, power and rate history, and, via equal-dimension state augmentation of the interference/fairness/distance blocks, the Lagrangian-bucket, delivery-debt (DVQ) and Dinkelbach-ratio blocks, plus split α, group id and QoS load/priority fields, and global time, SNR and CSI-error scalars — and a history window of the last H slots augments each user token. This is the feature-block space on which the interpretability analysis of §V-J operates. Three of the per-user blocks (arrivals, Lagrangian-bucket, Dinkelbach ratio) are *deterministic* at any fixed SNR evaluation point — zero variance there — and a pre-registered ablation (§V-J) shows they can be removed from the observation with no measurable effect; the pruned observation (10 per-user blocks, 531 dims at $H{=}4$) is therefore the new default configuration, while all results reported in this paper were obtained under the legacy unpruned observation for comparability with the champion checkpoint.

B. Hierarchical policy architecture

The meta-controller embeds per-user meta-features, applies multi-head self-attention, and outputs categorical group logits (temperature annealing $\tau : 2.0 \to 0.5$, balance regularizer), a group-level Gaussian head mapped by a sigmoid to $[\alpha_{\min}, \alpha_{\max}]$, and a mean-pooled critic. The worker is a self-attention graph network (SA-GAT): per-user tokens FiLM-modulated by global context with SNR and CSI-error scalars injected per token; 2 layers of 4-head attention (hidden 128) feed per-user heads for the $4K$ action and a mean-field critic pooling QoS class statistics.

C. Delivery-gated Lyapunov composite reward (DGLCR)

With served bits D_k, effective power P_{eff}, Dinkelbach ratio $q^\star$, delivery-debt queue Y and power-budget queue X:

$$\begin{aligned} r = c\Big[&V\big(\textstyle\sum_k D_k - q^\star P_{\text{eff}}\big) + w_{\text{tp}} \textstyle\sum_k \frac{Q_k}{Q_{\max}} w_k^{\text{qos}} D_k \\ &+ w_y \tfrac{Y}{Y_{\text{ref}}} \textstyle\sum_k D_k - w_q \sum_k \frac{Q_k^2}{Q_{\max}^2} \\ &- \textstyle\sum_k w_k^{\text{qos}} O_k - w_x \frac{X}{X_{\text{ref}}} P_{\text{tot}}\Big], \end{aligned} \quad (15)$$

where the first term is the Dinkelbach-linearized EE [15] ($q^\star$ on an outer loop), the second max-weight throughput, the third delivery-debt coupling, and the rest queue/overflow and power-budget penalties. Sigmoid gates on the EE term and

on penalties prevent the low-power trap $\partial \mathrm{EE}/\partial P < 0$ from dominating while delivery is infeasible. The Dinkelbach ratio is refreshed on an outer loop as

$$q^\star \leftarrow \frac{\mathbb{E}\big[\sum_k D_k\big]}{\mathbb{E}\big[P_{\mathrm{eff}}\big]}, \tag{16}$$

which is exactly the linearization that turns the fractional objective $\mathrm{EE} = R_{\mathrm{sum}}/P_{\mathrm{eff}}$ into the first term of (15).

D. Constraint stack

A Lagrangian dual variable λ on the delivery constraint (≥ 0.98) is updated with warmup and cap; its per-step signal is an EMA of instantaneous delivery. The EE policy-gradient is projected onto the null space of the delivery-constraint gradient (lexicographic projection). Per-user delivery virtual queues (DVQ) with CVaR tail focus (worst 25%) concentrate penalty on users risking the per-point gate. Concretely, with $c \geq 0$ the delivery-constraint residual and $\mathbf{g}$ the EE policy-gradient,

$$\begin{aligned} &\lambda \leftarrow \big[\lambda + \eta_\lambda(\tau - \widehat{\mathrm{deliv}})\big]_+^{\lambda_\cap}, \\ &\mathbf{g} \leftarrow \mathbf{g} - \frac{\mathbf{g}^\mathsf{T}\nabla c}{\|\nabla c\|^2}\,\nabla c \quad \text{if } c \geq 0,\ \mathbf{g}^\mathsf{T}\nabla c < 0, \end{aligned} \tag{17}$$

i.e. the dual ascent is warm-started and capped, and the projection removes the component of the EE gradient that would push the policy back into the delivery-infeasible region.

E. Solver expert and differentiable rate engine

Because $R(\cdot)$ is non-differentiable through π and the min-outage operator, we implement a torch shadow of the MMSE-SIC rate computation, verified float64-consistent with the immutable reference to atol 10^{-9}. Per slot, the expert runs a short Adam refinement (25 steps, lr 0.1, cold start) on power logits against a DGLCR surrogate mirroring the active reward branch, plus candidate-selection among decoding orders (gain-descending / queue-last). Formally, for each candidate order π_o the expert solves

$$\begin{aligned} &\theta_{\pi_o}^{(i+1)} = \theta_{\pi_o}^{(i)} - \eta_{\mathrm{Adam}}\,\nabla_\theta \hat{J}_{\mathrm{DGLCR}}(\theta;\pi_o), \quad i = 1,\dots,25, \\ &o^\star = \arg\max_o \hat{J}_{\mathrm{DGLCR}}(\theta_{\pi_o}^{(25)};\pi_o), \end{aligned} \tag{18}$$

cold-started every slot, where $\hat{J}_{\mathrm{DGLCR}}$ is the differentiable surrogate mirroring the active reward branch and θ are the power logits. Equation (18) contains no learned policy: it is a training-free, model-based per-slot optimizer whose only ingredient is the differentiable rate engine. The expert exceeds SDMA delivery at every tested SNR (0/5/15/30 dB: $+4.0/+1.4/+10.0/+23.9$ percentage points) at lower power.

F. Distillation and stepwise-frozen fine-tuning (SFTP)

Demonstrations from the expert (8,000 episodes) train the policy by behavior cloning (BC) on continuous heads and the Plackett-Luce head; BC alone clears all 7 gates. The policy is fine-tuned with PPO under SFTP: an anchor term injecting the BC gradient into the lexicographic-projection path, critic-only warm-up, a persistent anchor with slow decay, and guard-based checkpoint selection. Behavior cloning minimizes

$$\mathcal{L}_{\mathrm{BC}} = \mathbb{E}\big[\|\mathbf{a}_\theta^{\mathrm{cont}} - \mathbf{a}_{\mathrm{expert}}^{\mathrm{cont}}\|^2\big] - \mathbb{E}\big[\log P(\pi_{\mathrm{expert}} \mid \mathbf{z}_\theta)\big], \tag{19}$$

i.e. mean-square error on the continuous heads (power shares, drain ratios) plus the Plackett-Luce negative log-likelihood of the expert's decoding order, and the fine-tune optimizes the clipped surrogate

$$\begin{aligned} \mathcal{L}_{\mathrm{PPO}} = \mathbb{E}\big[\min\big(r_t(\theta)\hat{A}_t,\ \mathrm{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t\big)\big] \\ - c_v\mathcal{L}_{\mathrm{VF}} + c_e\mathcal{H}[\pi_\theta], \end{aligned} \tag{20}$$

with importance ratio $r_t(\theta) = \pi_\theta(a_t|s_t)/\pi_{\theta_{\mathrm{old}}}(a_t|s_t)$, GAE advantage $\hat{A}_t$, value loss $\mathcal{L}_{\mathrm{VF}}$ and entropy bonus $\mathcal{H}$. The combination of KL early-stopping and the learning-rate decay floor collapses the actor learning rate to its floor early ($\sim$update 100 of $\sim$780), an *emergent stepwise freeze* acting as an implicit trust region around the distilled basin; §V-D examines whether this freeze is causal.

Algorithm 1 Two-timescale hierarchical training with solver-expert distillation (SFTP)

```
Run the solver expert (18) for 8,000 episodes; store (s_t, a_exp^cont, π_exp)
Pre-train worker and meta policies by BC on L_BC (19)
Initialize λ, q*, delivery-debt queue Y, power-budget queue X
for episode = 1, ..., N do
    Meta: sample (g, α); hold fixed for T_s slots
    for slot t = 1, ..., T do
        Worker: sample (p, d, π) from π_θ(· | s_t) via (14)
        Env: rates (3), queues (7), reward (15)
    end for
    Update q* (16); update λ and project gradients (17)
    PPO update on L_PPO (20) with SFTP anchor and critic warm-up
end for
Select checkpoint by the per-point guard (6 SNR points × 16 episodes + queue bar)
```

G. Training protocol

Policy-gradient optimization via PPO [16], [17] with clip 0.1, rollout 2048, batch 64, 8 epochs, GAE $\lambda = 0.95$, actor lr 3×10^{-5}; a difficulty-annealing curriculum (5 stages) and hard-example mining re-weighting per-SNR-bucket sampling by delivery shortfall with streak multipliers and performance-linked entropy incentives. All mechanism flags default off so the official evaluation profile is unchanged. The complete pipeline is summarized in Algorithm 1.

IV. Experimental Methodology

Environment. $K = 24$, $N_r = 16$, $N_t = 2$, $T = 200$, $P_{k,\max} = 1\,\mathrm{W}$ (24 W aggregate), $Q_{\max} = 100$ bits; 3GPP UMi path loss; AR(1) $\rho = 0.9$; SNR-dependent CSI noise with $\varepsilon_{\mathrm{floor}} = 0.01$; unified MRT + MMSE-SIC; adaptive arrivals; 3 QoS classes.

Evaluation. Official sweep over $\mathcal{S} = \{0, 5, 10, 15, 20, 25, 30\}$ dB, 50 episodes/point, imperfect CSI; gates and lexicographic score per §II. Every score is produced by the same immutable evaluation script; checkpoints selected by the guard are re-evaluated officially.

Comparator set (classical + recent). Beyond classical SDMA [18], NOMA [19] and equal-power allocation [20], we compare against four recent (2025–2026) algorithms re-implemented in the identical environment and protocol: (i) *LDPTRLQ-style* single-timescale Lyapunov drift-plus-penalty RL in the family of Lyapunov-guided DRL for energy efficiency [9] (queue-stable DPP reward, no hierarchy, no distillation); (ii) *DR-SAC* [6], distributionally-robust SAC trained against the worst case over the SNR/CSI uncertainty domain (ICLR 2026); (iii) *neural-DPP*-style learned per-slot drift-plus-penalty control in the family of learning-augmented Lyapunov scheduling [11]; (iv) *DiffOpt*, a training-free model-based comparator running the differentiable MMSE-SIC shadow engine as a per-slot online optimizer (the solver expert without any learned policy), isolating learning from per-slot optimization. The repository's adaptive-power and water-filling heuristics are additional non-learned references. Trained comparators (i)–(iii) share the observation/action interface and are pre-registered before training; DiffOpt/adaptive/water-filling are evaluated on the identical grid (`scripts/evaluate_extended_baselines.py`).

Replication and pre-registration. The champion configuration is replicated verbatim at seeds 43 and 44 (besides seed 42). All fine-tuning variants were pre-registered (hypothesis, arms, single-variable controls, accept/reject gates) before launch; verdicts were written without post-hoc gate changes. Seed dispersion is reported alongside every comparison.

V. Results

A. Full-grid benchmark against classical baselines

Table I reports per-point delivery ratio (transmit power and mean queue parenthetically); Fig. 1 shows the comparison as curves with the seed-replication band (mean and min–max envelope over seeds 42–46). The policy clears all 7 per-point gates with score $[7,\ 0.9519,\ -8.3971,\ 1.0804]$: mean power 13.84 W ($<$ SDMA) and aggregate EE ratio 1.0804 (> 1.05). Training stability across the five seeds and the fine-tune/from-scratch-RL arms is shown in Fig. 2. The advantage is largest in the deep-overload high-SNR regime (30 dB: 0.888 vs. 0.664/0.581/0.601), and the policy uses roughly half the transmit power of fixed-power baselines.

B. From teacher to distilled policy

Pure behavior cloning on 8,000 expert demonstrations already reaches 7/7 gates with score $[7, 0.9441, -9.3929, 1.0753]$—the first configuration in this line to clear the full grid. SFTP fine-tuning improves it to the champion $[7, 0.9519, -8.3971, 1.0804]$ with zero guard rollbacks, i.e. a monotone-safe refinement of the distilled basin (Fig. 3).

C. Seed replication and the intrinsic noise floor

Verbatim replication at seeds 43–46 yields best-guard deliveries $\{0.9503, 0.9414, 0.9519, 0.9526\}$; with the champion (0.9519) the five draws give mean 0.9496, sd 0.0047, range 0.0112 (the pre-registered downgrade trigger, range > 0.012, is not met). The seed-44 best-guard (0.9414) is a checkpoint-selection artifact of the legacy 3-point mini-eval (its own final is 0.9491); seed 46's draw (0.9526) is statistically equivalent to the champion ($\Delta 0.0007$). End-of-training snapshots are markedly less stable than guard-selected checkpoints (finals n=4: mean 0.9362, sd 0.0200, vs best-guard sd 0.0047), retrospectively validating guard-based selection as the delivery artifact. Hence (i) the recipe centers at 0.9496 ± 0.0047 and the champion is one draw from that band, reproducible within ± 0.002; (ii) any comparison with margin $\lesssim 0.005$ is below seed resolution at single-seed sample sizes.

D. Pre-registered fine-tuning campaign

Five single-variable perturbations (Table II) all score 0.9446–0.9474 mean delivery—inside the champion's own seed band. The M5.3 arms additionally show their intended weak points *not* improving (E1: 0 dB 0.876, 30 dB 0.881, both below champion). We therefore report the fine-tune protocol's contribution as a seed-robust 0.9496 ± 0.0047 operating point and decline causal claims for individual mechanisms at present sample sizes; the pre-registered follow-on (3 freeze-onsets $\times$ 3 seeds) is the design that could resolve onset effects above the noise floor.

E. The model-based per-slot optimization upper bound

Table III adds the training-free comparators on the identical grid. Strikingly, *DiffOpt*—the differentiable MMSE-SIC engine run as a per-slot online optimizer (25 Adam steps on power logits plus decoding-order selection, *no learned policy*)—exceeds the champion policy at every SNR point (mean delivery 0.9574 vs. 0.9519, $+0.0055$, outside the ± 0.005 seed band; mean queue 7.36 vs. 8.40) while clearing all 7 gates. A causal variant whose surrogate uses only the estimated channel $\hat{\mathbf{H}}$ **matches the oracle-CSI variant point-for-point at 50 episodes** (max $|\Delta| \leq 0.001$ per point), so the advantage is not CSI leakage: even a fair, causal, training-free per-slot optimizer dominates the learned policy on this grid. Classical adaptive-power and water-filling heuristics remain far below both (30 dB: 0.777/0.778).

This re-frames the contribution hierarchy honestly: the differentiable engine plus solver expert is itself the strongest controller on this grid; behavior cloning compresses it into a single-forward-pass policy at a quantified cost (0.9574 $\rightarrow$ 0.9441), and SFTP fine-tuning recovers $\approx$59% of that compression gap ($\rightarrow$ 0.9519). The learned policy's remaining value is inference cost and model-free deployment (one forward pass per slot vs. 25 optimization steps with an explicit rate model), not absolute performance. Earlier "RL surpasses the teacher" wording is withdrawn: RL surpasses the *distilled student* (BC), not the live optimizer, which had not previously been evaluated on the official grid.

TABLE I
PER-POINT DELIVERY RATIO / POWER (W) / QUEUE; 50 EPISODES/POINT, IMPERFECT CSI.

SNR (dB)	**Ours (SFTP policy)**	SDMA	NOMA	EqualPower
0	**0.886**/13.72/2.84	0.856/14.69/3.46	0.836/17.35/4.01	0.791/24.00/4.94
5	**0.989**/14.39/0.43	0.979/13.70/0.68	0.970/17.35/0.92	0.943/24.00/1.53
10	**0.993**/14.59/0.69	0.961/15.02/2.28	0.947/17.35/3.10	0.915/24.00/4.62
15	**0.991**/13.71/1.91	0.887/15.12/11.42	0.839/17.35/15.54	0.824/24.00/15.48
20	**0.974**/13.39/7.20	0.795/14.62/22.39	0.715/17.35/27.40	0.726/24.00/25.50
25	**0.942**/13.59/16.08	0.724/14.64/28.76	0.636/17.35/33.07	0.658/24.00/31.80
30	**0.888**/13.52/29.63	0.664/14.87/34.47	0.581/17.35/36.83	0.601/24.00/37.27

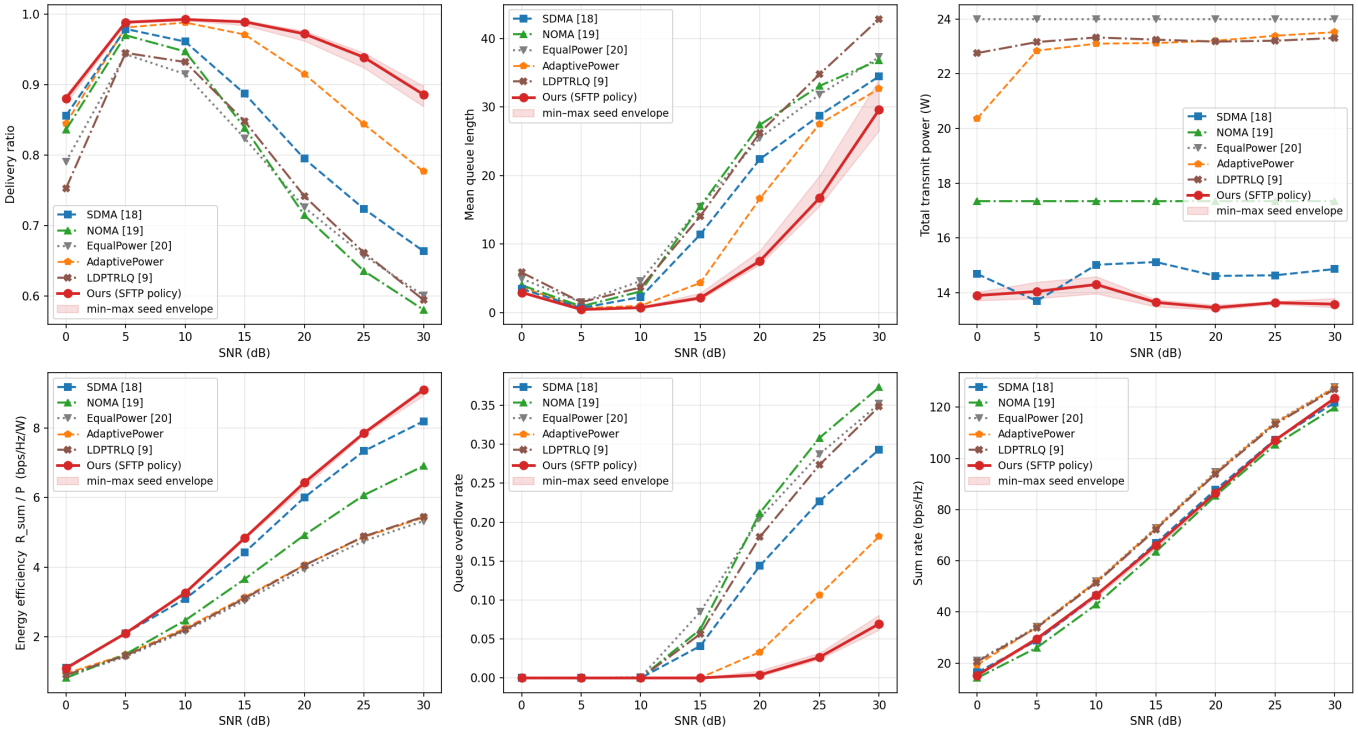


Fig. 1. Master benchmark. **Ours (SFTP policy)**: 5-seed mean with min–max envelope vs classical baselines (SDMA [18], NOMA [19], equal-power [20]), the adaptive-power heuristic (water-filling omitted: same channel-adaptive family, nearly identical delivery, they differ only in the power panel; tabulated in Table III), and from-scratch RL (LDPTRLQ [9]), on delivery, queue, power, EE, overflow and sum rate. Our engine's non-causal per-slot oracle optimizer (DiffOpt) and the classic-control arms (TD3/DDPG/JointOpt) are given as numeric comparators in Tables III and IV; individual seed members lie inside the envelope (§V-C). **Ours** dominates every plotted baseline at every SNR at roughly half the transmit power of the fixed-power schemes.

Trained recent-family comparator. A single-timescale Lyapunov drift-plus-penalty RL baseline in the LDPTRLQ family [9] (PPO with queue-Lyapunov penalties and the same curriculum/mining protocol, but no hierarchy, no distillation, no guard) clears **0/7 gates** (mean delivery 0.782, mean power 23.2 W $\approx$ saturated; Fig. 1). Vanilla queue-Lyapunov RL therefore cannot solve this overloaded instance from scratch; together with bc_init (7/7 from pure distillation) this localizes the enabling ingredient to *expert initialization*, not to the reward family or the hierarchy alone.

Classic continuous-control family (TD3/DDPG/SAC). The review-standard TD3/DDPG pair plus SAC, adapted to the mixed PL+continuous action space (deterministic pathwise actor through the differentiable $\tanh(\mu)$ continuous head and greedy PL order; twin-Q TD targets with target-policy smoothing and a delayed actor for TD3, single-Q for DDPG; single-timescale, no distillation, no guard): TD3 clears 0/7 gates (mean delivery 0.818 at a saturated 24 W transmit power), SAC also 0/7 (mean 0.787, its per-point queues exceed the gate at every SNR), and DDPG reaches 5/7 (mean 0.886) but still misses the two low-SNR gates $\{0, 5\}$ dB on both delivery and queue. All fall below the distilled policy (bc_init 7/7, 0.9441) and confirm the LDPTRLQ pattern: from-scratch RL cannot solve this overloaded instance, whatever the algorithm family.

Optimization-depth check (JointOpt). A per-slot optimizer with 60 warm-started Adam steps (vs. DiffOpt's 25 cold-start) scores below DiffOpt at every point (mean 0.913 vs. 0.957, $\approx$13 W): the model-based upper bound is not step- or warm-start-limited—cold-start 25-step DiffOpt is already tight, consistent with the earlier warm-vs-cold diagnostics.

Information-set vs. computation budget. The policy's deficit against DiffOpt could stem from (i) the oracle (true-channel)

TABLE II
PRE-REGISTERED FINE-TUNING VARIANTS (OFFICIAL BEST-GUARD SCORE).

Campaign / variant	Score	Verdict
Champion (SFTP, emergent freeze)	[7, 0.9519, −8.3971, 1.0804]	reference
M5.1 A2: actor lr ×3.3	[7, 0.9446, . . .]	lose
M5.1 A3: kill KL-decay starvation	[7, 0.9463, . . .]	lose
M5.2 D1: KL-to-BC trust region	[7, 0.9447, . . .]	lose
M5.2 D2: persistent strong anchor	[7, 0.9473, . . .]	lose
M5.3 E1: mining targets 0.92/0.93	[7, 0.9474, . . .]	lose
M5.3 E2: mining targets 0.95/0.95	[7, 0.9471, . . .]	lose

TABLE III
TRAINING-FREE COMPARATORS, PER-POINT DELIVERY (QUEUE), 50 EPISODES/POINT.

SNR (dB)	DiffOpt	DiffOpt causal	AdaptivePower	WaterFilling	Champion (ref)
0	**0.894** (2.64)	0.894 (2.64)	0.845 (3.75)	0.828 (4.12)	0.886 (2.84)
5	**0.990** (0.38)	0.990 (0.38)	0.981 (0.65)	0.979 (0.69)	0.989 (0.43)
10	**0.994** (0.62)	0.994 (0.62)	0.988 (1.01)	0.987 (1.07)	0.993 (0.69)
15	**0.992** (1.71)	0.992 (1.71)	0.971 (4.34)	0.969 (4.60)	0.991 (1.91)
20	**0.978** (6.50)	0.978 (6.49)	0.915 (16.63)	0.913 (16.65)	0.974 (7.20)
25	**0.949** (14.89)	0.949 (14.88)	0.844 (27.52)	0.845 (27.33)	0.942 (16.08)
30	**0.905** (24.75)	0.905 (24.74)	0.777 (32.67)	0.778 (32.63)	0.888 (29.63)

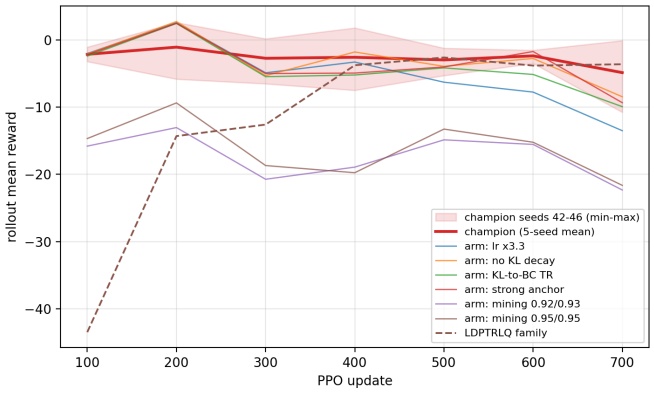


Fig. 2. Training stability (rollout mean reward vs. PPO update). Champion recipe over seeds 42–46 plotted as a 5-seed mean with min–max band; the pre-registered fine-tune arms (released lr / trust-region / anchoring / mining) stay within the band or below, and the from-scratch LDPTRLQ-family run collapses well below, consistent with the expert-initialization conclusion.

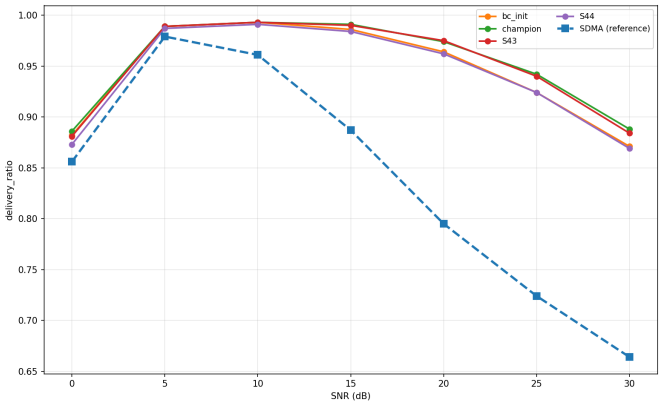


Fig. 3. Delivery ratio over the lineage: pure BC (bc_init), champion SFTP, and seed replications S43/S44; SDMA dashed reference. The seed band brackets the champion, confirming reproducibility within ±0.002 at seed 43.

information in the solver's surrogate or (ii) the per-slot optimization budget. Table IV separates both axes: the causal variant (surrogate restricted to the estimated channel $\hat{\mathbf{H}}$) reproduces the oracle variant point-for-point at 50 episodes ($\max|\Delta| \leq 0.001$), and the deeper 60-step warm-started JointOpt scores *below* the 25-step cold-start DiffOpt (0.913 vs. 0.957). The gap to the oracle is therefore intrinsic to replacing per-slot online optimization with a single shared forward pass—it is neither an information-set nor a step-budget artifact.

F. Checkpoint selection matters

The seed-44 episode exposes an evaluation pathology: a legacy 3-point mini-eval selected a checkpoint 0.9 pt below the same run's final model, because its saturated points cannot see mid-band regressions the official 7-point ×50-episode sweep reveals. A queue-aware 6-point guard (v2) catches these (v1 would have passed 5 bad checkpoints on the M5.2-D1 arm). Checkpoint selection must use the same per-point gate structure as the final evaluation.

G. Statistical assessment

Per-point deliveries are paired across the 7-SNR design grid (df = 6; two-sided paired t-test with Cohen's d; Holm–Bonferroni correction at $\alpha = 0.05$ across the eight external-family comparisons). The champion recipe's five-seed best-guard mean is 0.9496 ± 0.0047 (95% CI $[0.9438, 0.9555]$). **Ours** significantly exceeds SDMA ($p = 0.018$, $d = 1.22$), equal-power ($p = 0.004$, $d = 1.69$), water-filling and adaptive-power ($p = 0.016/0.023$, $d = 1.26/1.15$) and the from-scratch RL arms (LDPTRLQ/TD3/DDPG, $p \leq 0.006$, $d \geq 1.6$);

TABLE IV
CLASSIC-FAMILY AND OPTIMIZATION-DEPTH COMPARATORS (OFFICIAL 7×50, IMPERFECT CSI).

Comparator	points	mean deliv	30 dB deliv	P (W)
Ours (SFTP policy) [deliverable]	7/7	**0.9519**	0.888	13.5–14.6
bc_init (distillation, ours)	7/7	0.9441	0.871	13.7
DiffOpt / causal (oracle teacher upper bound, ours)†	7/7	0.957	0.905	13.4–14.6
JointOpt (60-step warm, ours)	5/7	0.913	0.872	≈13
DDPG [22]	5/7	0.886	0.792	13.6–21.3
SAC [31]	0/7	0.787	0.714	14.5–18.4
TD3 [21]	0/7	0.818	0.706	24 (sat.)
LDPTRLQ [9]	0/7	0.782	0.595	≈23

† Our solver-expert engine with oracle CSI and 25 Adam steps per slot; a non-causal reference, not a deployable policy.

TABLE V
ASYMPTOTIC AND MEASURED COMPLEXITY (PER SLOT UNLESS NOTED).

Operation	Asymptotic cost	Measured
Worker forward (SA-GAT)	$O(Kd^2{+}K^2{+}4Kh)$	≈19 ms
Meta (each T_s)	$O(Kd_m^2{+}K^2)$	negligible
PPO update	$O(BE(d^2{+}K))$	per rollout
DiffOpt (per slot)	$O(25N_r^2K{+}N_rK^2)$	≈0.26 s
DiffOpt saving	—	≈14×
Training (8,000 ep)	—	≈8 GPU-h/arm
Five-seed total	—	≈40 GPU-h

NOMA is non-significant after correction ($p=0.015$ raw, $d=1.28$). Against our engine's *oracle* per-slot optimizer the difference is small but detectable ($p=0.045$, $d=-0.96$): the single-pass causal policy sits a ≈0.6% compression gap below the oracle—the quantified cost of per-slot-online-optimization-free inference. Baselines are single-seed (seed 42); our recipe's multi-seed dispersion is reported in the seed-replication subsection.

H. Complexity and implementation cost

Table V lists asymptotic per-slot cost and measured runtime on the reference machine (8 GB GPU, 20-thread CPU). The worker is a 2-layer, 4-head self-attention graph network; a forward pass costs $O(K\,d^2 + K^2) + O(4K\,h)$ for the attention and per-user output heads, and the meta-controller adds $O(K\,d_m^2 + K^2)$ every T_s=20 slots. One PPO update is $O(B\,E\,(d^2 + K))$ with batch B=64 and E=8 epochs. DiffOpt spends 25 Adam steps on power logits plus two decoding-order candidates through the MMSE-SIC shadow engine, $O(25\,N_r^2K + N_rK^2)$, measured at ≈0.26 s/slot. The learned policy replaces this with a single ≈19 ms forward pass (rollout FPS 54, environment step included)—a ≈14× inference saving with no runtime dependence on the rate model. One 8,000-episode run takes ≈8 GPU-h; the five-seed report uses ≈40 GPU-h plus the CPU-only official evaluation grid.

I. Robustness to channel/traffic mis-specification

Table VI evaluates the champion under runtime-injected mis-specification (channel/traffic class patched at evaluation only; no retraining). Bursty two-state MMPP arrivals degrade mean delivery by only ≈0.02 while the margin over SDMA widens (+0.12 mean); LOS-rich Rician fading is the harder shift (−0.21 mean, as the receiver/precoder were tuned for Rayleigh), yet the policy still beats SDMA on average and at every point from 15 dB on; under the combined shift it matches SDMA at 0 dB and exceeds it elsewhere. The degradation is graceful and bounded—its magnitude, and the fact that the gap to SDMA narrows rather than inverts, quantify the policy's robustness to distribution shift without retraining.

J. Interpretability: what the policy attends to

Beyond aggregate delivery/EE numbers, we ask which *state features* drive the learned worker and meta policies, following two Shapley-value-based frameworks. **Global feature-subset selection via SVEA** [42]: we treat the 13 per-user observation blocks as players of a classification game whose characteristic function is the reduction in hinge-loss training error of a linear ERM classifier that predicts whether the policy serves a user in the current slot ($v(S) = \tilde{c} - \mathrm{tr_er}(S)$, $\tilde{c}$ the intercept-only error). The Shapley error-apportioning $e_j = \tilde{c}/n - \phi_j$ admits a universal threshold of zero: features with $e_j < 0$ (the *dominant* set SVEA_{neg}) are the ones whose knowledge alone reproduces the service decision. Fig. 4 reports e_j with 95% t-intervals per SNR point: the queue block is the sole dominant feature at 0–15 dB, split-α joins it at 20 dB, and no feature dominates at 30 dB (the policy serves near-uniformly and the balanced game finds no discriminative signal) — queue is the only feature whose dominance is consistent across points, overlapping the hand-selected mean-field critic channels but derived automatically (Table VII). The power of classification $PSV(\mathrm{SVEA}_{neg})$ (subset/full-set test accuracy) stays ≥ 0.95, i.e. the dominant features nearly reproduce the full 13-feature service model. The arrivals/λ-bucket/Dinkelbach-η blocks are constant within each fixed-SNR evaluation point and are excluded as null players (their marginal contribution is identically zero). We verified the practical consequence: a pre-registered same-budget retrain with those three blocks removed from the observation clears all 7 gates at mean delivery 0.9526 vs. 0.9527 for the 603-dim control ($\Delta=-0.0001$, one order below $\sigma_{\mathrm{seed}}\approx 0.005$) — removing them is empirically free, and the pruned observation (10 per-user blocks) is adopted as the default configuration. **Local attribution and reference sensitivity** [43]: for fixed states we compute expected-gradient attributions (model-introspective backpropagation against a reference) for the four

TABLE VI
PER-POINT DELIVERY UNDER RUNTIME MIS-SPECIFICATION (30 EPISODES/POINT, CHAMPION POLICY, NO RETRAINING).

SNR (dB)	nominal	Rician LOS	MMPP burst	Rician + MMPP	SDMA (both)
0	0.886	0.602	0.864	0.597	0.599
15	0.991	0.851	0.962	0.810	0.738
20	0.975	0.790	0.917	0.741	0.667
25	0.947	0.766	0.845	0.701	0.611
30	0.894	0.725	0.785	0.661	0.565
mean	0.939	0.747	0.875	0.702	0.636

decision objects — total/per-user power share, drain ratio, decoding-order confidence, and meta group/α — under four reference constructions (all-zero, per-block marginal means, empirical bootstrap, semantically neutral), and validate against the model-agnostic KernelSHAP proxy via top-5 recall and reference-pair Jaccard overlap (Fig. 5 and Table VIII). Mirroring the finding of [43], attributions are visibly reference-sensitive (reference-pair top-5 Jaccard often below 0.5), and the bootstrap/marginal-means references agree best with the model-agnostic proxy on the power target (recall@5 0.8–1.0 across SNR) while the all-zero reference is the least reliable — motivating the bootstrap reference as the paper's reporting default. KernelSHAP shows queue-backlog and channel-gain blocks as the dominant drivers of the total-power decision, consistent with the DGLCR design intent. Worker α is excluded as a dead dimension (overridden by the meta split mask).

The remaining three decision objects are *near-state-invariant* in the distilled policy: attributions for them are effectively zero, and direct rollout statistics confirm the mechanism rather than an artifact of the attribution. The drain ratio saturates at $d_k \equiv 1$ at every point ($\mathrm{std} < 10^{-4}$ across all 16,800 slots), the decoding order's top choice repeats only ≈0.15–0.19 of the time (vs. $1/24$ uniform), i.e. the softmax confidence on the top user is feature-flat even though the order is weakly structured, and the meta controller places essentially all users in a single group (≈1–2 groups per slot) with a near-constant split $\alpha \approx 0.15$–0.20 whose variance grows only at 30 dB. The power allocation is therefore the *only* strongly state-driven learned decision; the drain/order/grouping heads sit at learned default values, indicating that the SFTP distillation compressed the optimizer's power behavior into the policy and the other action heads kept their initialization-scale defaults — a concrete simplification opportunity for the architecture (fixed drain/order, or dropping the meta grouping).

VI. DISCUSSION

Why does the freeze help? The emergent early learning-rate collapse acts as an implicit trust region around the distilled basin, preventing drift into regions where the non-differentiable outage/SIC structure makes the surrogate reward misleadingly optimistic (notably 25–30 dB). Released-lr variants drift exactly there. Yet §V-D shows the frozen-vs-released difference is within seed noise, so the freeze is a stabilizer of a seed-robust operating point, not a proven causal lever; the multi-seed response surface is the correct instrument.

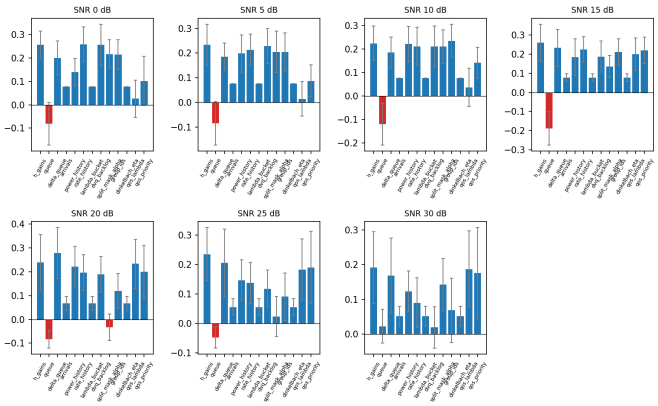


Fig. 4. SVEA per-feature scores e_j with 95% CIs per SNR point (red = SVEA$_{neg}$; constant features excluded as null players).

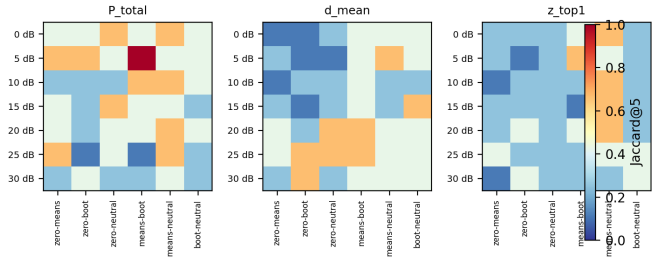


Fig. 5. Top-5 Jaccard overlap between expected-gradient attribution (reference variants) per target × SNR point.

The optimizer upper bound and what learning buys. The training-free per-slot optimizer (DiffOpt) exceeding the learned policy at every point forces an honest account: greedy model-based optimization with an accurate differentiable rate model is a stronger controller than any policy we trained, and the distillation+fine-tune pipeline is best understood as *compressing that optimizer into a fast, model-free policy* at a quantified cost (0.9574 → 0.9519, ≈1.2% mean delivery) in exchange for single-forward-pass inference and no runtime dependence on the rate model. The causal (estimate-only) variant retains the full advantage at 50 episodes, so the upper bound holds for fair causal controllers; the remaining open question is whether the distillation gap can be closed by distilling the optimizer's queue-conditioned behavior rather than its marginal actions.

Statistical honesty as a contribution. Single-seed RL com-

TABLE VII
PER-SNR SVEA$_{neg}$ DOMINANT SUBSETS: INTERVALS ARE 95% T-CIS OF THE GROUP-LEVEL e_j; PSV = SUBSET/FULL-SET TEST ACCURACY OF THE HINGE-LOSS CLASSIFIER.

SNR (dB)	SVEA$_{neg}$	e_j (95% CI)	PSV
0	queue	$-0.083 \ldots 0.008$	0.966
5	queue	$-0.004 \ldots 0.084$	0.953
10	queue	$-0.072 \ldots 0.017$	0.990
15	queue	$-0.087 \ldots 0.000$	1.003
20	queue,		
splitα	$-0.033 \ldots 0.005$ / $-0.019 \ldots 0.037$	1.008	
25	queue	$0.003 \ldots 0.039$	1.010
30	(none)	—	0.997

TABLE VIII
EG-VS-KERNELSHAP RECALL@5 ON THE POWER TARGET (P_{total}) PER REFERENCE CONSTRUCTION; ZERO-REFERENCE DEVIATES MOST, BOOTSTRAP/MARGINAL-MEANS REFERENCES AGREE BEST WITH THE MODEL-AGNOSTIC PROXY.

SNR (dB)	zero	means	bootstrap	neutral
0	0.6	0.8	0.8	0.8
5	0.6	1.0	0.8	0.8
10	0.4	0.6	0.8	0.8
15	0.4	0.6	0.8	0.6
20	0.8	0.8	1.0	0.6
25	0.8	0.8	0.8	0.8
30	0.6	0.8	0.6	1.0

parisons at effect sizes ~ 0.005 are indistinguishable from seed noise here. Pre-registered campaigns and explicit noise-floor quantification are intended as a template: report seed dispersion, pre-register gates, treat within-band differences as null.

Limitations. Fixed $K = 24, N_r = 16$; CPU-costly official evaluation limits replication; the expert surrogate inherits the reward's shaping choices; the emergent freeze is configuration-dependent and must be re-verified under reward or curriculum changes. The interpretability analysis §V-J is single-checkpoint and single-seed: the SVEA game uses a linear-surrogate ERM on the "served" outcome label, and the features constant within a fixed-SNR point (arrivals, Lagrangian-bucket, Dinkelbach ratio) carry no FSS information by construction; local attributions are reported under an empirical-bootstrap reference, chosen because it agrees best with the model-agnostic proxy, but the reference choice itself remains a judgment call, as documented in [43].

VII. CONCLUSION

We presented a hierarchical reinforcement-learning framework with solver-expert distillation for energy-efficient scheduling of overloaded RSMA MU-MIMO uplink under per-SNR delivery and queue constraints. The distilled, stepwise-frozen policy clears all seven evaluation gates with seed-robust mean delivery $\approx 0.950 \pm 0.005$ and dominates classical baselines across the SNR range at substantially lower transmit power. A pre-registered fine-tuning campaign establishes the seed-dispersion floor and sets multi-seed confidence-interval comparison as the reporting standard. Future work includes the multi-seed freeze-onset response surface and extending the solver expert to jointly refine the stream split, plus extending the interpretability analysis (§V-J) to per-user, per-slot attributions and to the queue-growth outcome label.

APPENDIX
KEY HYPERPARAMETERS

System: $K = 24$, $N_r = 16$, $N_t = 2$, $T = 200$, $P_{k,\max} = 1\,\mathrm{W}$, $Q_{\max} = 100$. CSI: AR(1) $\rho = 0.9$, $\varepsilon_{\mathrm{floor}} = 0.01$, robust reg. $\varepsilon^2 P_{\mathrm{tot}} \mathbf{I}$. Power: $\eta_{\mathrm{PA}} = 0.35$, $P_{\mathrm{circ}} = 5.0\,\mathrm{W}$, $P_{\mathrm{floor}} = 3.0\,\mathrm{W}$. Reward: $V = 5$, $w_{\mathrm{ee}} = 8$, $w_q = 2$, $w_{\mathrm{ovf}} = 20$, $w_y = 0.3$, $w_x = 1$, scale 0.01. Meta: $T_s = 20$, $G = 4$, $\tau : 2.0 \to 0.5$, $\alpha \in [0.1, 0.95]$. Worker: SA-GAT 2×(4-head, h=128), history $H = 4$. Observation pruning (A-prune, default on): arrivals / Lagrangian-bucket / Dinkelbach-ratio blocks removed (10 per-user blocks, 531 dims at H=4; champion evaluated under the legacy unpruned 603-dim observation). PPO: clip 0.1, rollout 2048, batch 64, 8 epochs, actor lr 3e-5, GAE 0.95. Expert: Adam 25 steps lr 0.1 cold-start, 8,000 demos. Guard v2: 6 points × 16 episodes + queue bar.

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